

Jayesh Limaye

A few months ago, we carried a comparison test of entry-level motherboards, which were meant for the budget-conscious consumer. This time round, we have a line-up of the choicest and best motherboards available in the Indian market—if you're ready to break your budget shackles in order to get the best.

In this comparison test of high-end motherboards, we're talking about top-of-the-line motherboards. Targeted primarily at PC enthusiasts, gamers, graphic designers and those who need to work with high-end applications such as video editing, these motherboards promise you performance to the core. Integrated with technologies such as infrared, Bluetooth, Wi-Fi, overclocking features, dual-core processor support and more, these motherboards carry a premium price tag.

The prices seem more justified because of the inclusion of high-end features such as SATA II, SLI, high-end audio, and advanced overclocking features in these motherboards. The inclusion of SATA II will support better storage capacities and performance. SLI dramatically improves graphics performance, which gamers will love. High-end audio means you can now really use your computer in your home theatre setup. Overclocking, again, is a gamer's delight—PC enthusiasts, too, will be able to play around with systems based on these boards and attain maxed-out performance.

To reiterate, systems in regular offices would not find a use for motherboards of this kind; the market here would be gamers and other enthusiasts. Built to cater to processors from market leaders Intel and AMD, these motherboards are a real joy to work upon.

We received a total of 20 motherboards—from ASUS, DFI, ECS, Foxconn, Gigabyte and MSI. Almost all these boards came with goodies (read lots of

cables and accessories) and packed boxes so attractive we wished we could have a separate photo shoot for these! As always, we gauged the motherboards based on features, performance and price, and zeroed in on the best of the best.

The boards were naturally divided into two categories—mobos for AMD and those for Intel. We talk about these in entirely different sections, as is appropriate.

MOTHERBOARDS (AMD)

In this comparison test, there were eight motherboards that supported AMD processors. Two of them were from ASUS, two were from Gigabyte, two were from MSI, and there was one each from DFI and ECS.

There happen to be four vendors who manufacture chipsets for AMD Athlon64 processors—ATI, nVidia, Via, SiS and ULI.

The various chipsets available for AMD motherboards are ATI's Radeon Xpress 200, nVidia's nForce4 Ultra and nForce4 SLI, SiS' SiS756, Via's K8T890, and ULI's M1573.

Technically speaking, the ATi Radeon Xpress 200 has the highest number of PCI-Express lanes—22—but two of these are utilised to connect the Northbridge with the Southbridge. nVidia and Via have 20 each, while SiS has only 16. This means all chipsets except SiS can support an x16 PCI-Express graphics card in addition to x4 slots.

While SiS756 supports an x16 PCIe graphics card, it has to rely on the Southbridge for additional PCIe slots. A thing worth noting is that of all the chipsets, only the nForce4 is a single-chip implementation, that is, there is no separate Northbridge and Southbridge.

Since the memory controllers for the Athlon64 class of processors reside on the CPU, the Northbridge does not have much to do, and the focus is more on the Southbridge, which has the storage, networking and audio controllers.

